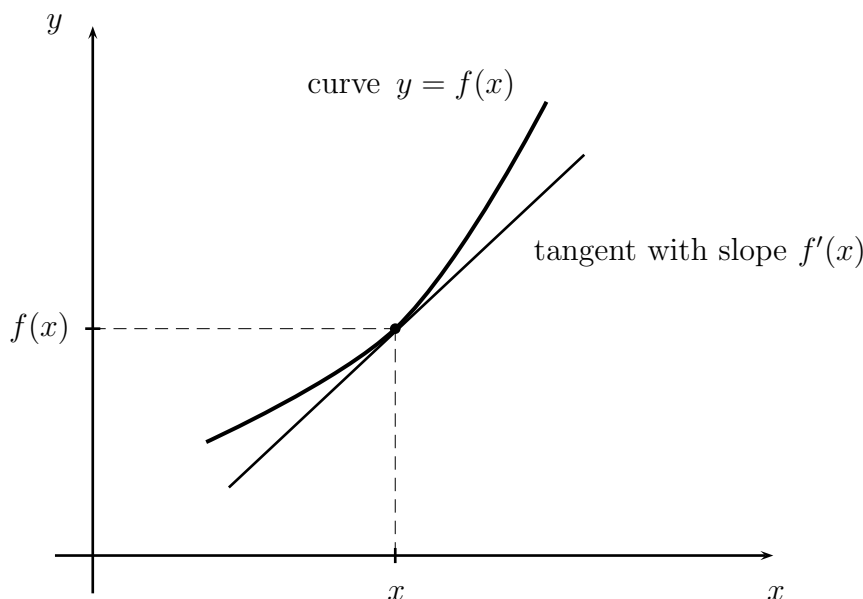

Notes for ‘The derivative as a limit’

Important Ideas and Useful Facts:

- (i) **Derivative of a function:** Consider a function $y = f(x)$. We define the *derivative* of f at x to be the following limit, if it exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Then $f'(x)$ represents the slope of the tangent line to the curve at the point $(x, f(x))$.



If $f(x)$ denotes displacement of an object at time x , then $f'(x)$ is the instantaneous velocity of the object at time x .

- (ii) **Derivative of the square function:** If $y = f(x) = x^2$, the rule for the square function, then $f'(x) = 2x$. For example, $f'(6) = 12$ and $f'(-7) = -14$.
- (iii) **Derivative of the cube function:** If $y = f(x) = x^3$, the rule for the cube function, then $f'(x) = 3x^2$. For example, $f'(6) = 3(6)^2 = 108$ and $f'(-7) = 3(-7)^2 = 147$.
- (iv) **Derivative of the power function:** If $y = f(x) = x^n$, the rule for the power function with fixed exponent n then

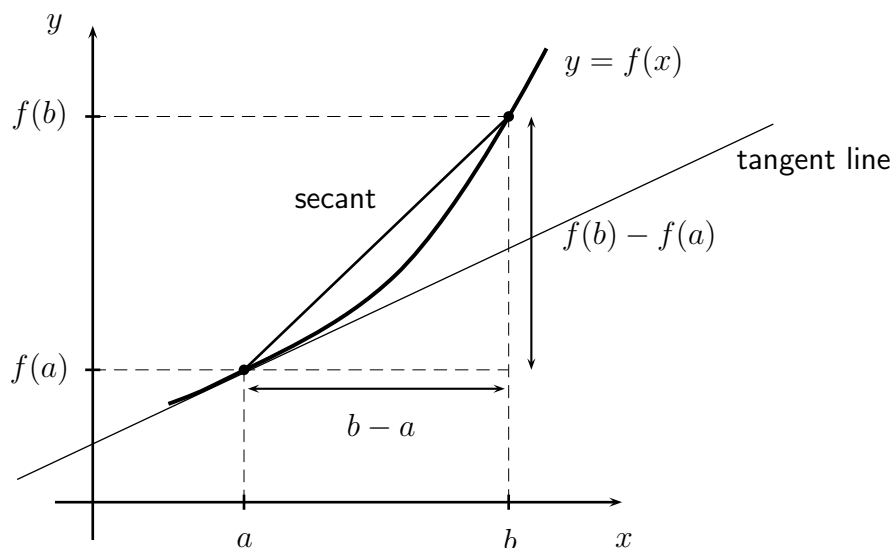
$$f'(x) = nx^{n-1},$$

that is, the exponent falls down to the front, and the new exponent is obtained by subtracting one. The derivatives of the square and cube functions are special cases when $n = 2$ and 3 .

For example if $f(x) = x^4$, then $f'(x) = 4x^3$. If $f(x) = x^{-3}$ then $f'(x) = -3x^{-4}$. If $f(x) = x^{1/3}$ then $f'(x) = \frac{1}{3}x^{-2/3}$.

This rule works generally for all real exponents (though there may be certain restrictions such as requiring x to be positive). An explanation why this works so generally will be given in the next module.

- (v) **Alternative limit definition of the derivative:** An alternative, but equivalent, approach to finding an expression for the slope of the tangent line to the curve $y = f(x)$ is to take the average rate of change over an interval $[a, b]$, where $a < b$, and allow b to approach a .



The slope of the tangent line at $x = a$ then becomes the derivative

$$f'(a) = \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a}.$$

To check algebraically that the limit definitions really are equivalent, put $h = b - a$, so that $b = a + h = x + h$, and then this becomes

$$f'(x) = f'(a) = \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a} = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h},$$

noting that $h \rightarrow 0$ as $b \rightarrow a$.

Suppose that $f(x) = x^3$. We verify that $f'(x) = 3x^2$, using both limit definitions.

Using the original definition, we have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x + h)^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} \\ &= \lim_{h \rightarrow 0} \frac{3x^2h + 3xh^2 + h^3}{h} = \lim_{h \rightarrow 0} \frac{(3x^2 + 3xh + h^2)h}{h} = \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) \\ &= 3x^2 + 3x(0) + (0)^2 = 3x^2, \end{aligned}$$

as required. Alternatively, to practise the second definition, based on the limit of the average rate of change, we have, using a difference of two cubes formula in the middle step,

$$\begin{aligned} f'(a) &= \lim_{b \rightarrow a} \frac{f(b) - f(a)}{b - a} = \lim_{b \rightarrow a} \frac{b^3 - a^3}{b - a} = \lim_{b \rightarrow a} \frac{(b - a)(b^2 + ba + a^2)}{b - a} \\ &= \lim_{b \rightarrow a} (b^2 + ba + a^2) = a^2 + a^2 + a^2 = 3a^2, \end{aligned}$$

again giving $f'(x) = 3x^2$, by taking $x = a$.